

Phase 7: Project Documentation

OptiCrop: Smart Agricultural Production Optimization

Role	Name
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Setup & Installation

1. Ensure Python 3.8+ is installed on your system.
2. Open a terminal in the root **OptiCrop** directory.
3. (Optional but recommended) Create a virtual environment: `python -m venv venv` and activate it.
4. Install required dependencies using the command: `pip install -r requirements.txt`

Running the Application

1. **Model Generation:** Open the **Training/** folder and run `model_training.ipynb` in Jupyter Notebook. Execute all cells to generate `crop_model.pkl` and `scaler.pkl`.
2. **Move Models:** Copy the two generated `.pkl` files from the **Training/** folder into the **Flask/** folder.
3. **Start Server:** Navigate to the **Flask/** directory in your terminal and run: `python app.py`
4. Access the application by opening your web browser and going to `http://127.0.0.1:5000`

User Guidelines

- **Navigation:** Use the green top navigation bar to switch between the Home, About, and Find Your Crop pages.
- **Making a Prediction:** Go to "Find Your Crop". Enter the numerical values for Nitrogen, Phosphorous, Potassium, Temperature, Humidity, pH level, and Rainfall into the respective input boxes.
- **Viewing Results:** Click the "Predict" button. If successful, a green box will appear below the form displaying the recommended crop name. If an error occurs (e.g., missing fields), an error message will be shown instead.